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Threat Register

Subject V.6 requires more than protections: *"you must identify other dangers and describe the protection measures that can be implemented"*. This is that document.

A written accepted risk is an engineering decision. An unwritten one is an oversight. Evaluators ask what you did *not* protect against; this table is the answer.

Status

Design-stage register. 27 threats. Slice S0 landed the first verified rows, S1 added T-25, T-26 and T-27, and closed T-25; every other mitigation below is *designed*, not verified. The Verified column stays empty until a test, a config artifact or a named code path proves it.

Legend

Likelihood and Impact: L / M / H. Residual = what remains **after** the planned mitigation.

Register

#	Threat	L	I	Planned mitigation	Residual	Accepted by	Verified
T-01	Credential stuffing / brute force on login	H	H	rate limit per account **and** per IP, progressive backoff, generic errors, BCrypt cost >= 12	**mitigated 2026-09-27 by SPEC-006**, within the limits of one instance: distributed low-and-slow attempts across many IPs remain possible, the per-account counter is the control that bounds them, and there is no breached-password feed at launch. A restart clears every counter	owner, 2026-09-27, on SPEC-006	S1: `CredentialRateLimitingT` `CredentialAttemptLimite` and `BCryptPasswordHasherTes` for the cost floor
T-02	Session hijacking via stolen access token	M	H	15-minute access token, HTTPS only, no token in URLs or logs	a stolen token is valid until expiry; no device binding		—
T-03	Refresh-token theft	M	H	opaque, hashed at rest, rotated on every use, family revoked on reuse detection	detection is after the fact, not preventive		—

#	Threat	L	I	Planned mitigation	Residual	Accepted by	Verified
T-04	Broken object-level authorization (another user's data)	M	H	two gates in a pipeline behaviour, checked on every request and hub join; hidden resources return `404`	a new endpoint that bypasses the behaviour; mitigated by a required negative test per endpoint		—
T-05	Vote manipulation (double voting, stuffing)	H	M	unique index `(EventId, TrackId, UserId)` as the arbiter, rate limit on the vote endpoint	multiple accounts by one person (sybil) is not prevented; email verification raises the cost only		—
T-06	Playlist reorder conflict causing lost or divergent order	H	M	sparse ordering keys plus concurrency token, reload-and-reapply or `409`	rapid concurrent edits still surface as `409` to the user		—
T-07	Account takeover through unverified provider email matching	L	H	match an existing account only when the provider asserts `email_verified`	a provider that wrongly asserts verification		—
T-08	ID token substitution (token minted for another app)	L	H	validate `aud`, `iss`, `exp`, `nonce` against JWKS, server-side	depends on correct JWKS caching and key rotation handling		—

#	Threat	L	I	Planned mitigation	Residual	Accepted by	Verified
T-09	API key or secret committed to git	L	**critical**	`.env` git-ignored from commit 1, `.env.example` placeholders only, rotate-first leak procedure	subject V.8 makes this an **automatic project failure**; process discipline is the only control		S0: `MusicRoom/.dockerignore` excludes `.env`; no `.env` and no credential string in any layer of the built image; `docker-compose.yml` carries no literal
T-10	YouTube quota exhaustion during the demo	M	M	prefer `videos.list` over `search.list`, debounce, cache `TrackRef`, handle `403 quotaExceeded` explicitly	a determined user can still burn quota; no per-user quota budget at launch		—
T-11	Denial of service via unbounded input or list requests	M	M	capped `pageSize`, bounded string and collection lengths, global plus per-endpoint rate limits	a single instance has no upstream protection; capacity is whatever V.7 measures		—
T-12	SignalR connection exhaustion	M	M	bounded payloads, explicit keep-alive and timeout settings	no backplane (issue O8), so one instance is the ceiling		—
T-13	Database connection-pool exhaustion	M	M	small pool, short-lived connections, pooled connection string for the API	Supabase enforces a hard connection ceiling above ours		—

#	Threat	L	I	Planned mitigation	Residual	Accepted by	Verified
T-14	Personal data over-exposure in profile responses	M	M	per-field filtering server-side by requester relation-ship	a new profile field added without a visibility tier; mitigated by making the tier non-nullable		—
T-15	Sensitive data in logs	M	H	never-logged list, structured templates, no full bodies on credential endpoints	a developer logging an object graph that contains a token		partial, S0: `GetHealthReady_WhenData` proves the readiness body carries no host, connection string or exception text
T-16	Vulnerable transitive depen-dency	M	M	`dotnet list package --vulnerable --include-transitive` fails CI on high severity	a zero-day between scans		S0: step present in both workflows and in `make agent-check`; clean across all 10 projects
T-17	Location+time licence bypass (spoofed GPS)	M	M	server-side geofence check **plus** an optional short-lived event code shown at the venue; the licence policy selects which proofs are required — [[ADR-0014-location-time-licence]]	client-reported coordi-nates remain spoofable; the event code raises the cost but a code can be relayed to someone off-site	`UNKNOWN - requires decision`	—

#	Threat	L	I	Planned mitigation	Residual	Accepted by	Verified
T-18	Malicious content in user-supplied text (event names, playlist titles)	M	M	validation, length caps, output encoding at the consumer	the client is responsible for rendering safely; no web frontend today		—

#	Threat	L	I	Planned mitigation	Residual	Accepted by	Verified
T-19	Mail provider quota exhausted by a registration loop	L	M	rate limits on registration and verification resend; hosted provider free tier — [[ADR-0013-mail-provider]]	a quota exhausted mid-defence blocks the verification demo. **The rate limiter is a precondition of slice S1a** (owner decision, SPEC-004, 2026-09-27): `POST /api/v1/auth/verification-requests` is the first endpoint whose success path is one mail send per anonymous request, so the moment O17 lands the hosted provider the loop stops being theoretical. While `LoggingMailService` is the only sender the cost is log lines on a Development host, which the SPEC-003 start-up guard confines to that environment		—

#	Threat	L	I	Planned mitigation	Residual	Accepted by	Verified
T-20	Supabase free project pauses from inactivity	M	H	touch the project before the defence; local PostgreSQL container as a fallback with identical migrations — [[ADR-0016-supabase-plan-region]]	a pause discovered on the day still costs demo time		—
T-22	Malicious controller skips the queue	L	M	only `controllerUserId` may advance; `expectedNowPlayingId` makes a replay idempotent; the owner can revoke control at any time	a legitimate controller can still skip a song — that is what control means		—
T-23	Playback stalls because the controller device went offline	M	M	control takeover via `PUT /devices/{id}/control`; the owner can revoke	no server-side watchdog by design ([[ADR-0017-playback-session]]); recovery is manual (issue O13)		—

#	Threat	L	I	Planned mitigation	Residual	Accepted by	Verified
T-24	Operational endpoints reachable by anyone who can reach the port	M	L	`/health` and `/health/ready` are anonymous by contract ([[endpoint-inventory]]) and report only check names and states — no version, host, connection string or exception text	an unauthenticated caller learns the service is up and whether its database is reachable, and can poll it: the endpoints carry no rate-limit policy, because a limiter that throttles a probe defeats the probe		S0: readiness body asserted by test; no authentication is planned for these two

#	Threat	L	I	Planned mitigation	Residual	Accepted by	Verified
T-25	Anonymous auth endpoints have no rate limit: `POST /api/v1/auth/registrations`, `POST /api/v1/auth/email-verifications`, `POST /api/v1/auth/verification-requests`, `POST /api/v1/auth/sessions` and `POST /api/v1/auth/tokens` can all be driven in a loop	M	M	rate limits keyed on IP and on the submitted address (SDD §13.3), built once for registration, verification, login and resend together	**closed 2026-09-27 by SPEC-006.** Deferred three times while the control was shared with endpoints that did not exist; built once, for all five at once. Each endpoint carries its own per-IP policy, so a registration loop cannot spend the sign-in allowance, and login, registration and resend also carry a per-account counter with progressive backoff. Refusals are `429` with `Retry-After` and an empty body, so the refusal itself says nothing about which key tripped. **Residual, carried deliberately:** limiter	owner, 2026-09-27, on SPEC-006	S1: `CredentialRateLimitingT` — per-IP limit with `Retry-After`, per-account limit scoped to one address, a second client unaffected, and the registration, resend, verification and rotation loops each bounded; `CredentialAttemptLimite` — the window, the doubling penalty, the ceiling and the reset on success

#	Threat	L	I	Planned mitigation	Residual	Accepted by	Verified
T-26	Development-only `LoggingMailService` writes the email verification link, token included, to the application log	M	M	registered as `IMailService` only when the host environment is Development; `AddInfrastructureServices` throws `InvalidOperationException` at start-up in any other environment rather than falling back silently or booting with no sender. With no `VERIFICATION_LINK_BASE_URL` configured the sender logs the bare token and says the variable is missing — it never invents a default link	**accepted.** Anyone who can read a developer's container logs can verify an account created on that machine. The exposure is bounded to Development hosts by a mechanical guard, not a convention, and a production host fails loudly instead of leaking. The value is that the O17 gap does not block the flow: a developer reads the link out of `docker compose logs` and exercises verification end to end before a mail provider exists. Removed when O17 lands the hosted provider (slice S1a), at which point this row closes	owner, 2026-09-26, on SPEC-003	S1: `LoggingMailServiceTests` asserts one event carrying the link, and the non-Development start-up throw is a named code path in `InfrastructureServiceRe`

#	Threat	L	I	Planned mitigation	Residual	Accepted by	Verified
T-27	No security event is logged when a credential endpoint refuses a request: a failed login, a rejected refresh token and a detected refresh-token **reuse** all leave no record beyond the Serilog request line	M	M	the A09 event list in [[owasp-top10-checklist]]: log failed login, lockout, token reuse, authorisation denial and rate-limit trip, through the audit behaviour of slice S2 ([[ADR-0019-hub-action-auditing]], SDD §13.4)	**partly mitigated.** SPEC-006 logs the **rate-limit trip**, with the endpoint and the wait and nothing that identifies the account, so a guessing run is now visible. Still unlogged: an individual failed login, a rejected refresh token and a detected family revocation — the theft detector can still fire with no trace an operator could investigate. Those three are audit records, they touch every mutating endpoint, and they land with slice S2	accepted for S1 by SPEC-005, narrowed by SPEC-006, owner 2026-09-27 — closes in slice S2, which precedes the defence	S1: the `OnRejected` warning in `CredentialRateLimiterPo`

#	Threat	L	I	Planned mitigation	Residual	Accepted by	Verified
T-21	Voter location data stored unnecessarily	L	M	the submitted coordinate is used for the check and **not** persisted on the vote row; event centre stored at coarse precision	the coordinate exists in transit and in request logs unless excluded — add it to the never-logged list		—

Explicitly out of scope

Not defended against	Why
A determined attacker with unlimited resources	subject V.6: *"your API is not expected to be unbreakable"*
Physical device compromise	the token is in the platform secure store; a rooted device is outside our control
Supabase platform compromise	vendor trust boundary, stated rather than pretended away
Sybil accounts (T-05)	email verification raises cost; full prevention needs identity proofing this project will not build

Maintenance

- A finding from [[owasp-top10-checklist]] or [[owasp-api-top10-checklist]] that is not fixed in the pull request gets a row here **before** that PR merges.
- Fill Accepted by with a real name when the team agrees to carry a residual risk.
- Fill Verified with the test or artifact that proves the mitigation, once it exists.
- Never delete a row. Close it by pointing at the evidence.

Related

[[owasp-index]] · [[asvs-conformance]] · [[application-security]] · [[authentication-authorization]] · [[sdd]] §13.6 · [[subject-requirements]]